



FACULTY INSTRUCTORS

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Course site: I'll send you links/instructions

COURSE TIME AND PLACE

This class meets:
MWF: 11:00AM – 11:50 pm (3064)
Thursday 1:00PM – 3:50 PM (3064)

See notes on meeting and lab time below

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Office hours:
M, W, 8-9AM, and M 3-4PM

I. COURSE INFORMATION

Term and Year: FALL 2026

Course Prefix, Number, and Title: PHYS, 4010, Advanced Laboratory

Number of Credit Hours: 3

Course Description: Experiments are conducted in various fields of physics including modern physics and optics. Evaluation, analysis and interpretation of experimental data is emphasized.

Prerequisite(s): PHYS 3300 (C or better) or permission of the instructor.

Department Name: Department of Physics and Biophysics

College: College of Science and Mathematics

Midterm Last Day for Withdrawal, OCT 21.

II. GRADING AND ASSIGNMENT DESCRIPTIONS

COURSE EXPECTATIONS

WHAT IS EXPECTED FROM YOU

General

- Arrive to lab on time.
- Be prepared for your lab activities (read handouts and background ahead of time)
- Be proactive in understanding what is needed to do well in lab. Ask questions when you don't understand something.
- Bring all needed materials to each lab: Lab notebook, calculator, etc.

- Arrive on-time with a can-do attitude. There will be times when things don't work the first time (this is the real world of lab so get used to it); keeping a positive outlook and avoiding frustration are important for a good lab experience.
- In order to complete each lab you will need to come to lectures, listen carefully, and read lab handouts beforehand. You will be provided with support videos for many (maybe not all) of the labs
- Leave your area neat, clean and organized (for the next group).
- Comment on meetings
 - Times for course will be used for either lectures on background material or course logistics or lab acquisition demonstrations.
 - The most important times will be those "new lab intro and demo".
 - We have many items for course logistics (discussing HW, handing in HW, peer review, etc). Some of the times this might be used for YOUR lab data acquisition time. -. You will be signing up for lab data acquisition times.
 - In general you are expected to perform lab /data acquisition on your own time (I am arranging for room access 24/7). In some cases we will not use full times on M,W,F or the long Thursday lab sessions. These are times you should be available and should general or Wed. times for lectures/demos/other. During such times and with my permission, you may (if needed) schedule your lab times during these blocks of time on Thursday, or to include some of the MWF time if we are not using that. Since we have ONE SETUP—You will typically schedule your own lab data acquisition time (with your rotating partner). There will be a signup sheet.

Experiments

- Read all lab handouts before coming to lab. You must consider the physical principles involved with each lab experiment. Your notes on lab procedures and data acquisition steps should be in your lab notebook before you come to the lab.
- Complete your assignments and submit them on time.

WHAT YOU CAN EXPECT FROM ME

- Arrive to lab on time, plan the course and make adjustments as needed.
- Give you feedback – both written and oral. Each lab assignment in this should be considered the rough equivalent of an exam in most courses. I will give feedback.
- Specifics on handing in labs and course components will be detailed on my website.

COURSE GRADES

Your final course grade stems from the following grading opportunities below:

Course grades will be assigned as follows:

A = 90 – 100%; **B** = 80 – 90%; **C** = 70 – 80%; **D** = 60 –70%; **F** = <60%

ASSIGNMENT DESCRIPTIONS:

A. LABS (several components)

For each lab you will complete data acquisition with a partner (in general I will assign partners). The lab work will be graded based on three components.

- **Data Analysis 20% of course grade. This may be thought of as a rough draft.**

A data analysis component, where you are expected to have completed all data acquisition, all graphs and numerical analysis including uncertainty. What you hand in

must contain a clear set of data, theory and math used to analyze, and clear description. A reader must be able to take their own data set (raw) and apply what you have presented to yield the end result. The result must be interpreted by you.

This will typically be due one week after the end of data taking. You will hand in a neat edited document (no handwriting, and use common professional notation---no “E” like 3.14E6, but instead, 3.14×10^6 and no 10^6 either---no, no, no, no). This should have length needed to complete the analysis and any graphs you intend to include in your final draft.

- Collected data (with uncertainty) (never use +-, but always $\pm$ for uncertainty).
- Graphs
- Fitting curves (with uncertainty)
- Calculations (ask if not sure---I don’t want to see remedial math steps)
- Comparison with the theoretical, known, or standard value (when possible)
- You don’t need to write up detailed procedure or produce experimental figures until the final draft, but I should be able to tell what you did (start writing).
- I do need to know basic procedures (enough to follow how you got your numerical results).

Grades will be based on clarity, neatness, correctness, uncertainty, significant figures, properties of plots or graphs, and any other items needed to determine results. This should be well organized, so anyone can explain your calculations just following your write up. This work will be written using LaTeX or other document preparation tools/software ---keep it neat.

I’ll be providing a rubric to help guide you. It is a guideline (no hard fast rulebook).

- **Final Report 35% of course grade**

The final report is written as a research styled paper. This includes an abstract, properly formatted writing, references, analysis, graphs, figures, experimental description and setup, captions, tables, etc--- Here you are writing as if you did this lab first as research. This will be the final draft of your lab report. It will have the style and quality of a research paper, like for example AJP (American Journal of Physics) which is the same as the American Physical Society template. You will use LaTeX (or other document prep—I still use Word). You must have a thorough understanding of all components of the lab, and clearly communicate the lab procedures and any relevant technical issues relevant to the performance of the experiments. You must thoroughly interpret and discuss your results. More detail on my course materials. Don’t ask me “Am I done?” or “Have I done everything you want?---there is no upper limit, and my answer will be to think of more that can be done. It’s up to you to decide “sufficient”, likewise with the experimental part. There are typically some MAIN POINTS in the experiment (don’t miss that).

- **Oral lab review and lab notebook 10 % of the course grade**

For each lab you will have a scheduled session with me on your own, with your lab notebook (no floppy stuff). I will personally mess up the setup, and you will have a limited time (typically 30 minutes) to take some sample data and do a rough analysis as requested (typically on the whiteboard). This may use any features of the instrumentation and require understanding of any of the material within background for a given experiment. I may also inspect your lab notebooks during these sessions. You are expected to have complete, accurate, legible and well maintained lab notebooks—so that you can get through this part of the process. Lab notebooks contain notes, contents,

handouts, data, notes to yourself on how to analyze and use the instrument. Don't underestimate the power of thewell maintained lab notebook.

The lab notebook is the component that says you were in lab and performed the experiment. Your lab notebooks should contain all the information you need to enter lab and understand how to complete data acquisition, as well as your rough analysis. You should leave several pages for contents to start, and about 8 or so pages for each lab. Have the lab title, and each of the sections discussed below (a,b,c,d) in your lab notebook. Your lab notebooks will be checked for:

- a. notes regarding "today's lab" (indicates you are prepared for today's lab).
 - i. These are your handwritten notes prior to the start of lab.
 - ii. This section is not the handout ---which you may have externally/separately with you in lab.
- b. Your data for today's lab
 - i. In general if your data is in a file---include the file name /date/time and where you store it
 - ii. If you know you will make a graph/plot---leave room for it in this part.
- c. If possible there should be a rough check on "today's result" before you are done with lab (so check your numbers before you walk out the door).
- d. Your rough analysis from last week (if you make a plot or perform a calculation to obtain a result, include it in your lab notebook).
- e. It's fine to include lab handouts in your notebook (do so). Just don't confuse my instructions with your own "legwork" and instructions to yourself.

- **Peer Review of labs 5% of course grade**

You will be assigned a paper with names stripped off/redacted. You will complete a comprehensive (blind—you won't know whose) review of the papers and present this orally to the class during one of our meetings (document camera). Your goal is to give honest, useful constructive criticism of the paper you are looking at. The end result should be advice to the author for ways to improve the paper. We will have open conversations about whether the paper should be making changes or whether the reviewer has misunderstood or just has an opinion. You will be graded based upon whether you have done a diligent review. A diligent reviewer will have credible suggestions to improve the paper. Ultimately, if the reader is diligent, then the Author is the only person responsible for possible mis-readings or misunderstandings. It is always the task of the Author to clear up any possible mis-readings or misunderstandings. There are issues relating to style that should be much less significant than understanding.

B. Above and Beyond/Projects (20% of the course grade)

As you complete each lab you are building and using skills that you develop in this course and or bring from other experiences. You are expected to make strong connections in your own way between course material and your own skillset. You will complete a course project (typically with a partner) on a topic. This may be a significant extension of labs we have done, or a new project of your own. I'll give a list of some suggested project ideas. I must ensure the project is doable with equipment we have or can purchase. I do expect innovation and creativity.

As part of grading I will prompt early **proposals** (though you may look ahead at labs not done yet). You may have more than one potential idea. In addition I will prompt you to

complete and demonstrate your project to my satisfaction (it must be at least as substantial as any of the labs we have done—you are creating the writeup, description, analysis etc). The results of your project will be presented to faculty ---my choice—as either oral power point, or poster. Be prepared.

C. HOMEWORK: (10% course)

There will be various assignments throughout the course on topics such as error analysis, curve fitting, or specific support work for specific experiments. I may give pre-lab assignments, sample data analysis, to help you get started on some of the lab understanding.

Late work will generally not be accepted without a confirmed non-academic valid emergency. Missing an assignment deadline may result in zero credit for that assignment. You must have all components of a given lab completed (analysis must be completed on time and certainly before the final report). Always discuss with me if there is a problem! It is always OK to ask me!

While it is expected to work with lab partners, ultimately the writing and production of figures will be your own. Handing in plagiarized work may result in penalty of zero on all lab components for that lab (and has happened before). Equivalent penalty for all collaborators (don't give your work to others). No AI writing.

Lab experiments

1. Speed of Light
2. Schawlow's Ruler
3. Michelson Interferometer
4. Zeeman Effect and Bohr magneton
5. Hydrogen/Deuterium (newish) and Sodium Spectra
6. Planck's Constant

III. LAB SCHEDULE

TENTATIVE COURSE SCHEDULE

Week		Lab Start	Comments
1	August 18 (Tuesday start)	Speed of light - Intro Wed. Aug. 19. Data done by Aug. 24 (end of day) - Analysis draft Due by Friday Aug. 28th in class - Final Draft Sept. 4th in class	
2	Aug. 24		Individual Oral Reviews will be scheduled during Analysis due week. (between August 24 th and 28 th) Your responsibility.
3	Aug. 31	Schawlow Ruler Start – - Data Start on Wed. Aug. 26th (Demo/Intro). Data done Sept. 4 - Analysis Due Sept. 11 - Final Sept. 18th	Division of wavefront PROJECT PROPOSALS SHALL BEGIN
4	Sept. 7.		I expect a fit of the non-linear “exact” description. We’ll go over this. Oral-lab review during this week
5	Sept. 14	Michelson <i>Intro MONDAY Sept. 8th</i> Data done Sept. 18th - Analysis Due Sept. 25th - Final due Oct 2nd	Division of Amplitude BEGIN PEER REVIEWS DURING M,T,W TIMES

6	Sept. 21		Oral -lab review
7	Sept. 28	Zeeman • Intro Sept. 25th • Data done Oct. 2nd • Analysis Due Oct. 9th • Final Due Oct. 16th	PROJECT BRAINSTORMING CONTINUES
8	Oct. 5		Oral Lab Review
9	Oct. 12	Hydrogen • Intro Oct. 12th • Data done by Oct. 16th • Analysis Oct. 23th • Final Due Oct. 30th	
10	Oct. 19		Hydrogen Oral
11	Oct. 26	Planck and Photoelectric • Intro March 24th • Data Done by Oct. 30th • Analysis Nov. 6 • Final Due Nov. 13	
12	Nov. 2	Project Time	Planck Oral
13	Nov. 9	Project Time	
14	Nov. 16	PRESENT NOV 20TH	
15	Nov. 23		

Lab groups/lab partner information will be determined during the first week of class.

IV. POLICIES

ATTENDANCE POLICY

If you are unable to attend classes you should communicate with your instructors. You are responsible for all material, assignments, etc. I will consider any missed assignments, lab or any class activity as an absence. Excessive absence or tardiness may result in your being withdrawn or receiving a failing grade in the course. Ultimately it is the student responsibility to make sure that the proper paperwork has been handed in by midterm (or later on, if that is the case) in order to ensure withdrawal (or WF). Generally 10% of the course is considered too much to miss, and it is possible you will be withdrawn.

Make ups for work missed will be scheduled individually, but only for acceptable reasons. You must notify me beforehand if possible.

In addition to the above-mentioned instructor policy, you are also obligated to follow the Augusta University Attendance Policy. This university policy essentially states that you are expected to punctually attend all classes from the first to the last day of the term, that your instructor will monitor both your attendance and participation, and that there are academic penalties for excessive absenteeism.

ACADEMIC INTEGRITY

You are encouraged to study together and to discuss information and concepts covered in the lecture and the lab sessions with other students. You can give “consulting” help or receive “consulting” help from such students. However this permissible cooperation should never involve one student having possession of a copy of all or part of the work done by someone else, in the form of an e-mail, an e-mail attachment file, a picture, a flash drive, a hard copy, or code

etc. For example you may share a matlab graph viewing with another student, but not the code---not line by line, etc.

It is the duty of the student to practice and preserve academic honesty. Each student should be aware of the specific policies governing academic conduct for the course, as well as the grievance and appeals processes put in place for adjudicating such policies. If the student has any doubt about a situation, he or she should consult with his or her instructor. It is also the student's responsibility to maintain his/her correct address of record with Augusta University so that official notification of the student regarding academic misconduct can be carried out in timely fashion.

Violations of academic integrity may range from penalties on graded work such as receiving a grade of zero for particular assignments, or may result in receiving WF in the course. You should consider lab reports in this course, each the equivalent of a major exam—so any violations of academic integrity on any component of lab work is a serious violation!

DISORDERLY CONDUCT

Augusta University prohibits behavior that disrupts the academic, research or service mission or activities of the University, or disrupts any activity or event of the University community. Some examples of disorderly conduct include, but are not limited to, the following: conduct which causes a breach of the peace; lewd, obscene or indecent conduct; conduct which interferes with or disrupts activities or functions sponsored or participated in by the University or by members of the University community; conduct that is disruptive to a classroom lecture, lab, or other teaching or research entity of the University, interfering with or obstructing pedestrian or vehicular traffic; obstructing or interfering with ingress or egress of campus buildings or facilities; conduct which interferes with the rights of others; unauthorized use of electronic or other devices to make an audio or video record of any person without his or her expressed or implied consent when such recording is likely to cause injury or distress.

MORE COURSE INFORMATION

General Learning Outcomes

- Learn to employ high scientific standards in written work as demonstrated in lab reports.
- Students will interpret experimental work from other papers (peer review)
- Students will give oral presentations (peer review and projects).
- Practice clearly and concisely articulate scientific ideas and arguments through written works.

Specific Course Objectives

- Students will use experimental equipment such as lockin amplifier, oscilloscopes, and spectrometers.
- Students will interpret and explain experimentally observed phenomenon
- Analyze data and make conclusions about physical phenomena

V. DISCLAIMER

RESERVATION TO CHANGE THE course outline or syllabus

The course instructor reserves the right to make changes to the course information and schedule with reasonable notice to the students.